AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

***Code*: MT129**

Tutor-Marked Assignment (TMA) Fall 26/27

Cut-Off Date: Based on the Published Deadline.

Total Marks: …. marks turned to 10 marks

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Plagiarism Warning:

As per AOU rules and regulations, all students are required to submit their own TMA work and avoid plagiarism. The AOU has implemented sophisticated techniques for plagiarism detection. You must provide all references in case you use and quote another person's work in your TMA. You will be penalized for any act of plagiarism as per the AOU's rules and regulations.

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I hereby declare that this submitted TMA work is a result of my own efforts and I have not plagiarized any other person's work. I have provided all references of information that I have used and quoted in my TMA work.

Name of Student:……..

Signature:……...

Date:……

MT129-TMA

[A] Student Component

Student Name : _____

Student Number : _____

Group Number : _____

[B] Tutor Component

	Comments	Weight	Mark
Q_1		10	
Q_2		10	
Q_3		10	
Q_4		10	
		40	

Comments:

Do not use colors or highlight in writing the answers. Use the same file to write the answers (Don't use PDF file). Type your answers (insert $\rightarrow Equation$).

Please solve each question in the space provided. You should give the details of your solutions and not just the final answer.

Q–1: [2+3+1+4 marks]

a) Find the domain of the following functions:

i) $f(x) = \frac{1}{\sqrt{7x}}$

ii) $g(x) = \frac{2}{x^2+3x}$

iii) $h(x) = x^3 + 3x^2 - 4x$

b) Find the points of intersection of the following two curves:

$y = 30x^3 - 3x^2$ and $y = 16x^3 + 25x^2$

Q–2: [4+2+4 marks]

a) Use the definition of the derivative to compute the derivative for the function $f(x) = \sqrt{x-2}$.

b) Let $f(x) = x^5$ and $g(x) = 6x - 1$, find $\frac{d}{dx} f(g(x))$.

c) Given the function $y = (x^5 - (x-1)^5)^{10}$, find $\frac{dy}{dx}$.

Q−3: [5+5 marks] let $f(x) = x^3 - 6x^2 + 9x + 3$

a) Find the intervals on which $f(x)$ is increasing or decreasing and find the local maximum and local minimum, if any.

b) Find the intervals on which the graph of $f(x)$ is concave up or concave down, and find the inflection points, if any.

Q−4: [3+4+3 marks]

a) Find an equation of the tangent line, at the designated point: $\sqrt{x} + \sqrt{y} = 7$, p(9, 16).

b) Using logarithmic differentiation to find y' if $y = \ln \frac{(x+1)^4}{e^x - 1}$.

c) Solve the equation: $2(\ln x)^2 + \ln x - 1 = 0$.